Computer Networks Lab
Week 01

Introduction to Packet Tracer and Network Topology Setup

Tools:

Cisco Packet Tracer 8.2+: Download from https://www.netacad.com/resources/lab-downloads?courseLang=en-US

Objectives:

1. Understand the concept of IP addressing and subnetting.
2. Navigate the Cisco Packet Tracer interface proficiently.
3. Design and configure a basic LAN topology using routers, switches, and PCs.
4. Use the CLI to configure devices.
5. Troubleshoot connectivity and common physical/logical configuration errors.

Introduction to Networking

A **computer network** is a collection of interconnected devices that communicate with each other to share resources and data. Networks play a vital role in modern computing by enabling collaboration, communication, and resource sharing, such as accessing files, printers, and internet services.

Networks can range from small **Local Area Networks (LANs)**, which connect devices within a single building, to vast **Wide Area Networks (WANs)** that span across cities, countries, or even continents. The primary purpose of these networks is to ensure communication between devices and optimize resource usage.

As networks grow in complexity, learning how to design, configure, and troubleshoot them is essential. To achieve this, network simulation tools like **Cisco Packet Tracer** are invaluable for building and understanding networks in a controlled, virtual environment.

IPv4 Addressing Fundamentals

Understanding IP addressing is crucial for network configuration. IPv4 addresses are organized into classes, each serving different network sizes and purposes.

IPv4 Five Classes

Address Class	1st Octet range in decimal	1st Octet bits (Blue Dots do not change)	Network (N) and Host (H) Portion	Default mask (Decimal)	Number of possible networks and hosts per network
A	0–127	00000000 - 01111111	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts (2^{24}–2)
B	128–191	10000000 - 10111111	N.N.H.H	255.255.0.0	16,384 Nets (2^{14}) 65,534 hosts (2^{16}–2)
C	192–223	11000000 - 11011111	N.N.N.H	255.255.255.0	2,09,150 Nets (2^{21}) 254 hosts (2^8–2)

Important IPv4 Concepts:

Class D and E:

- **Class D** (Range: 224.0.0.0 to 239.255.255.255) - Reserved for multicast
- **Class E** (Range: 240.0.0.0 to 255.255.255.255) - Reserved for experimental use

Private IP Addresses:

- Reserved specifically for private/internal use only
- Cannot be routed on the public Internet
- Used for internal networks (home, office, etc.)

Subnet Mask:

- 32-bit number created by setting host bits to all 0s and setting network bits to all 1s
- Identifies the network portion and host portion of an IP address
- Determines which devices are on the same network

Gateway IP:

- A network node used in telecommunications
- Serves as an entry and exit point for a network
- All data must pass through or communicate with the gateway prior to being routed
- Usually the router's IP address on that network

Internet

The **Internet** is the world's largest network, connecting millions of devices globally through standardized **protocols**. Protocols are rules that define how devices communicate with each other across networks.

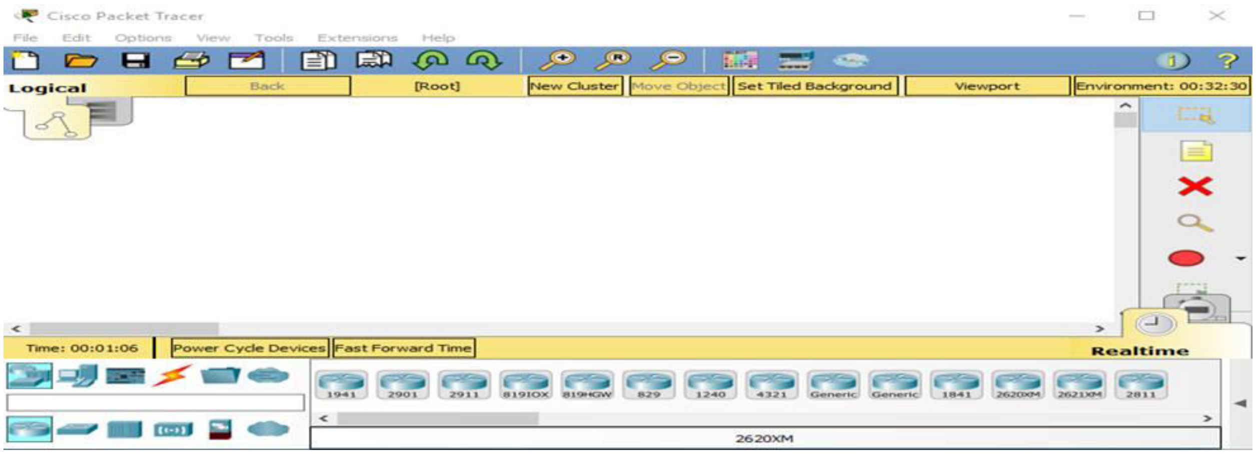
Common Internet Protocols:

- **IP (Internet Protocol)**: Addressing and routing data packets
- **HTTP (Hypertext Transfer Protocol)**: Web page transfer
- **SMTP (Simple Mail Transfer Protocol)**: Email transmission
- **FTP (File Transfer Protocol)**: File sharing
- **TCP (Transmission Control Protocol):** Reliable data delivery
- **UDP (User Datagram Protocol):** Fast, unreliable data delivery

Key concepts:

- **Network Edge:** End systems that run applications
- **Access Networks:** Connect end users to Internet (DSL, Cable, WiFi)
- **Network Core:** Routers that forward packets between networks
- **Physical Media:** Guided (cables) vs Unguided (wireless)

Introduction to Cisco Packet Tracer



Cisco Packet Tracer's interface.

Cisco Packet Tracer is a powerful, interactive network simulation tool developed by Cisco Systems. It provides a hands-on platform for designing network topologies, configuring devices, and troubleshooting issues - all without needing physical hardware.

In this lab, you'll learn how to use Cisco Packet Tracer to create and manage a basic network topology, helping you understand foundational networking principles while gaining practical skills with this versatile tool.

Exploring the Cisco Packet Tracer Interface

When you open Cisco Packet Tracer, you will see a user-friendly interface with various menus, tools, and workspaces. Here's a brief overview of the key components:

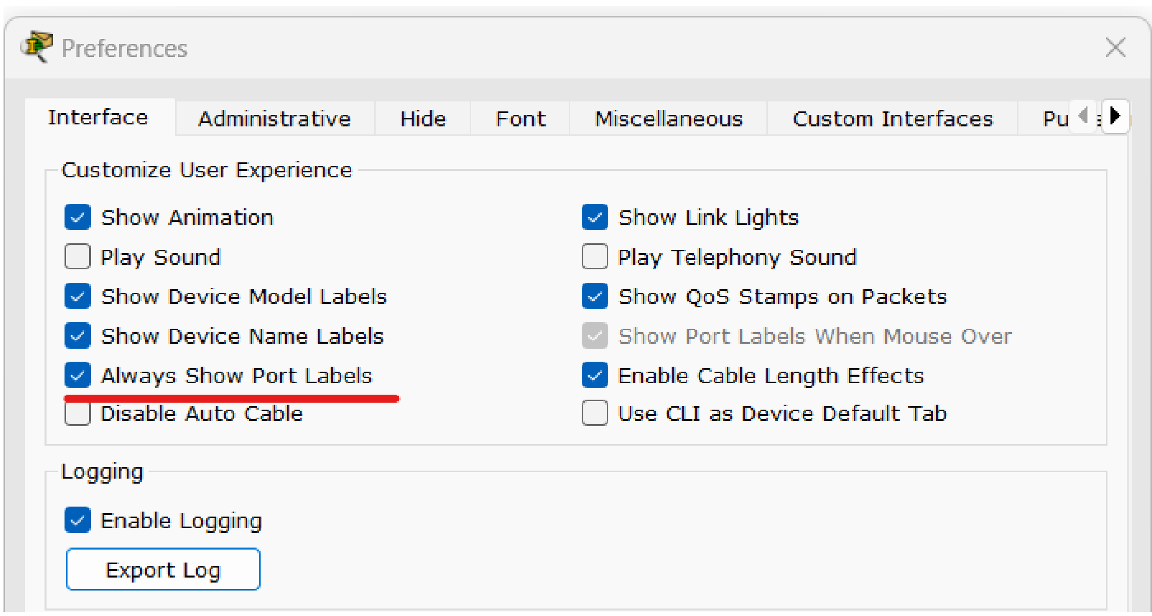
1. **Menus and Toolbars**:
    - **a.** The top menu bar provides access to file operations, device selection, and simulation controls.
    - **b.** The toolbar contains shortcuts to commonly used tools, such as selecting devices, connecting cables, and deleting objects.
2. **Workspaces**:
    - **a. Logical Workspace**: This is where you design and simulate your network topology. You can drag and drop devices, connect them, and configure their settings.
    - **b. Physical Workspace:** This allows you to visualize the physical layout of your network, such as where devices are placed in a building or data center.
3. **Simulation Mode**:

a. Packet Tracer offers a simulation mode where you can observe how data packets travel across the network. This is useful for understanding how devices communicate and troubleshoot issues.

Essential PT Settings:

Enable Port Labels for better visibility:

1. Click on **Options** → **Preferences** or **Ctrl+R**
2. Enable the following options:
   - **Show Device Model Labels**
   - **Show Device Name Labels**
   - **Always Show Port Labels**



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Devices to be Used in Labs:

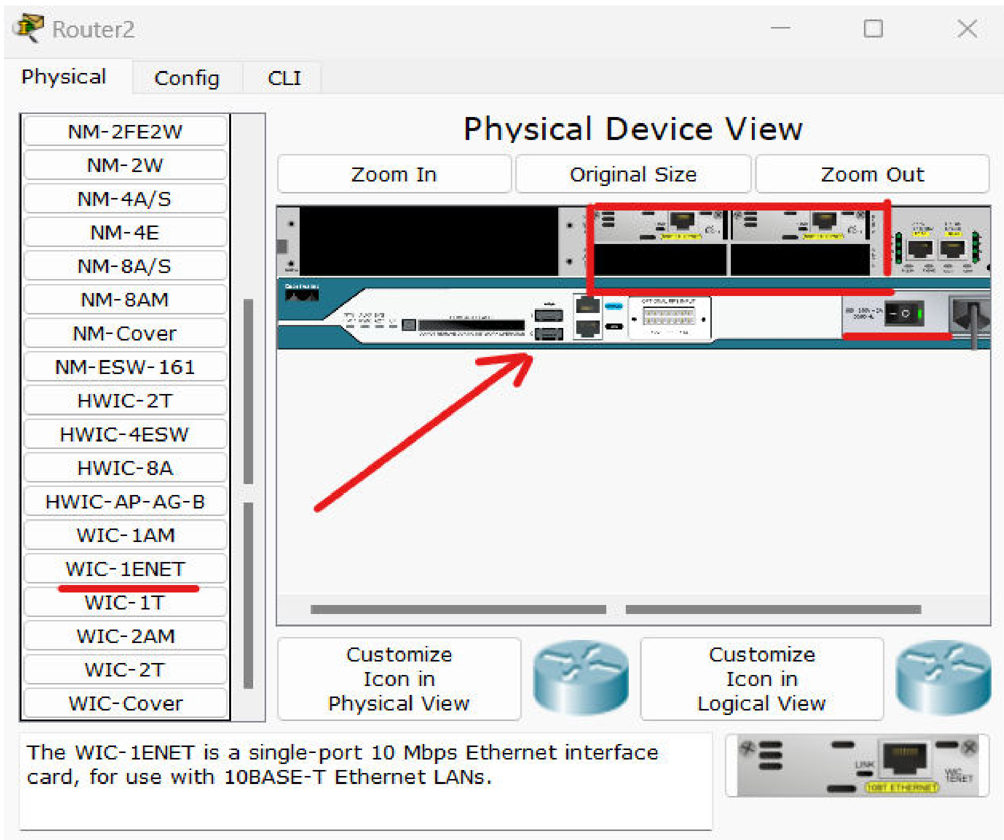
Standard Equipment:

1. **Routers**: 2811 (or 1841 for basic labs)
2. **Switches**: 2960-24TT (or simple 2960)
3. **End Devices**: Generic (any PC or laptop)

Connection Types:

- **Copper Crossover**: connecting similar devices (PC to PC, Switch to Switch)

- **Copper Straight-through**: connecting non-similar devices (PC to Switch, Switch to Router)
- **Serial DTE**: between routers for WAN connections



Adding Network Interface Cards to Routers:

Default Router Interfaces: By default, routers have two ethernet connections: **Fa0/0** and **Fa0/1**. To add more interfaces:

Step-by-Step Process:

1. **Click on router** to open device window
2. **Switch off router** by clicking the button with green light (power button)
3. **Drag the interface card** from the left panel into an available slot
4. **Turn on router** by clicking the power button again
5. **Make connections** using appropriate cables to the new interfaces

Available Interface Cards:

- **WIC-1ENET**: Adds one additional FastEthernet port

- **WIC-2T**: Adds two serial ports for WAN connections
- **NM-4E**: Adds four additional Ethernet ports

Physical Media in Packet Tracer

Packet Tracer includes various cable types representing different physical media:

1. **Copper Straight-Through**: Standard Ethernet cable (twisted pair)
2. **Copper Cross-Over**: For connecting similar devices
3. **Fiber Optic**: High-speed optical connections
4. **Serial**: For WAN connections between routers
5. **Phone**: For telephone/DSL connections

Choosing the Right Cable:

- **PC to Switch**: Straight-through copper
- **Switch to Router**: Straight-through copper
- **PC to PC (direct)**: Cross-over copper
- **Long distance/high speed**: Fiber optic

Basic Device Configuration

Router Configuration Using CLI:

Step-by-Step Router Setup:

1. Click on router
2. Go to **CLI** tab
3. Write **'no'** if option for continuing with configuration dialog is coming
4. Enter the following commands for interface configuration:

```
Router> enable
Router# configure terminal
Router(config)# interface fastEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
```

Note: In the command "ip address 192.168.1.1 255.255.255.0":

- **192.168.1.1** is the gateway IP address for this network
- **255.255.255.0** is the subnet mask

- Change these values according to your network design

PC Configuration:

Assigning IP to PCs:

1. Click on PC (e.g., PC0)
2. Select **"Desktop"** tab
3. Click on **"IP Configuration"**
4. Configure the following:
   i. **IP Address**: Write IP address for PC from network range (e.g., 192.168.1.2)
   ii. **Subnet Mask**: Enter appropriate subnet mask (e.g., 255.255.255.0)
   iii. **Default Gateway**: Enter router's IP address (e.g., 192.168.1.1)
5. Close the configuration window

Important: Ensure all devices in the same network use:

- Same network ID (first three octets for Class C)
- Same subnet mask
- Correct gateway IP (router's interface IP)

Creating a Simple Network Topology

A network topology refers to the arrangement of devices and connections in a network. In this lab, you will create a simple topology consisting of PCs connected to a switch.

Network Design Principles:

- **Network Edge**: PCs representing end systems (clients)
- **Local Network**: Switch connecting devices in same network
- **Network Core**: Router representing connection to other networks (if needed)
- **Physical Media**: Appropriate cables for each connection type

Step-by-Step Setup:

- **Add End Devices (Network Edge)**:
  - Drag and drop PCs from the End Devices menu onto the logical workspace
  - These represent hosts at the network edge
- **Add Access Network Device**:
  - Drag 1 Switch (2960-24TT) from Network Devices
  - This provides local network access
- **Connect with Physical Media**:
  - Connect each PC to the switch using **Copper Straight-Through** cables
  - Connect to different switch ports (FastEthernet0, FastEthernet1, etc.)

- **Configure IP Addresses**:
  - Assign IPs from the same subnet (e.g., 192.168.1.x/24)
  - Use consistent subnet mask (255.255.255.0 for Class C)
  - Configure gateway if router is present

Verifying Connectivity

Once the network is set up, you need to verify that the devices can communicate with each other.

Using the ping Command:

- Open the command prompt on one PC and use the ping command to test connectivity with the other PC
- Example: ping 192.168.1.11
- A successful ping indicates that the devices are correctly connected and can communicate

Commands to Use:

```
PC> ping 192.168.1.11     (Test connectivity)
PC> arp -a                (View ARP table)
PC> ipconfig              (View IP configuration)
```

Documenting the Network

Proper documentation is essential for understanding, maintaining, and troubleshooting a network. In this lab, you will document your network topology using labels and notes.

To annotate the topology:

1. Use labels to identify each device (e.g., PC1, PC2, Switch1)
   a. Go to **Options** → **Preferences** → **Tick the boxes** for 'Show Device Model Labels' and 'Device Name Labels'
2. Add notes to specify the IP addresses assigned to each device
3. Document cable types used for each connection
4. Create a network diagram showing IP addressing scheme

Lab Tasks

Task 1: Explore the Cisco Packet Tracer Interface (5 Marks)

Explore the Cisco Packet Tracer interface by opening the software and navigating through its menus, tools, and workspaces to become familiar with its layout and functionality. Focus on identifying key features such as the **Simulation Mode**, which allows you to visualize network packet flow, and the **Logical Workspace**, where you design and configure network topologies.

Additionally, examine the **Physical Workspace**, which provides a real-world representation of network setups. Then, take clear screenshots along with your roll no. on the screen of the interface and label the following key components: Menu Bar, Tool Bar, Device Specific Selection Box, Workspaces (Logical and Physical) and Annotation and Labels .

Task 2 & 3: Create a Simple Network Topology and verify connectivity (10 Marks)

Part A - T2 (5 marks)

Create a network topology in Cisco Packet Tracer by adding three PCs, one switch and one router to the workspace.

Manually configure IPv4 addresses for the devices; assign **192.168.1.5** to PC1, **192.168.1.6** to PC2, and **192.168.1.7** to PC3 and ensuring they are within the same subnet. Use the **ping** command from PC1 to test connectivity with PC2 and so on to verify successful communication.

Part B - T3 (5 marks)

Introduce a common issue, such as using a crossover cable instead of a straight-through cable or assigning mismatched IP addresses that fall outside the same subnet. Document the troubleshooting process, including identifying the issue, testing configurations, and correcting it by replacing the cable or reconfiguring the IP addresses, and then retesting connectivity to confirm the resolution.

Task 4: Documenting the Network (5 Marks)

Annotate the network topology created in Tasks 2 and 3 by adding labels and notes to identify each device and its corresponding IP address. Clearly label the devices as **PC1**, **PC2, PC3**, **Switch** and **Router**, and indicate the IP addresses (e.g., **192.168.1.5** and **192.168.1.6**) and cable types used. Take a screenshot of the fully annotated topology.

Also write a concise description of the network, explaining its purpose, such as demonstrating basic connectivity and troubleshooting. Include a summary of each device's role, the type of connections used, and how the setup enables communication between the devices.

Submission Guidelines:

- Submit a single PDF document containing all tasks and the CISCO packet files to the GCR, **ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **J_i221234_Lab1**.
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **J_i221234_T1**.

During a demo, there will be deductions based on your answers to the questions asked.

Demo:

- If student cannot answer at all -5 marks.
- If student can answer somewhat -2 marks.
- Correct answers results in no deductions.

NOTE: Screenshots where your name, roll number and section are NOT visible, will automatically be given 0.

Sample screenshot: